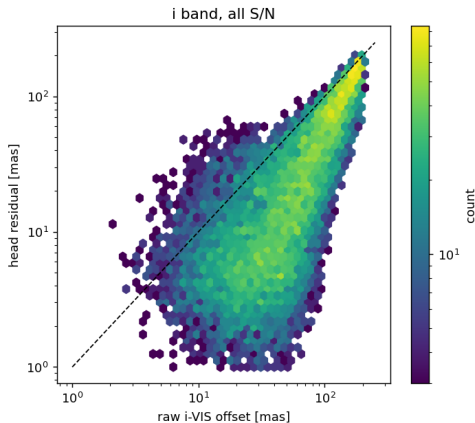
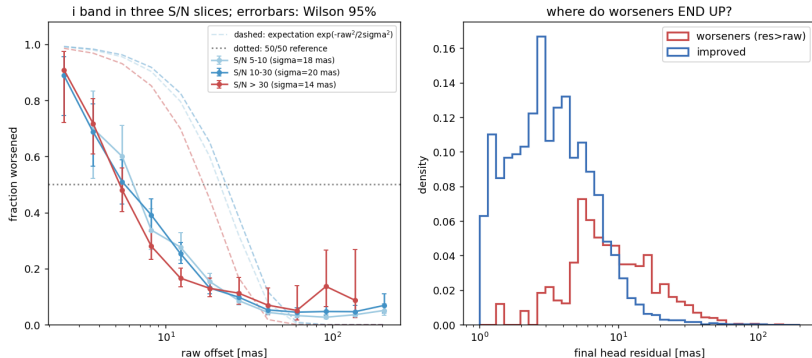


Why Astrometry Is Degraded for a Subset of Objects: Diagnosis and Cure in JAISP

July 2026, updated August and September 2026 (internal, ECDFS Q1)



- The head (**v10 production**) improves the population (raw ~ 48 mas $\rightarrow$ 13.4 mas median, all bands; panel: i band), but 9% of all measurements end up **farther** from VIS than they started.
- Two candidate explanations: (a) those sources were lucky before, and the small raw offset was chance; (b) the model fails on those sources specifically.



v10 production head

- Left: the same pattern in every S/N slice, each observed curve below its own re-measurement expectation (dashed, $P(\text{worse}) = e^{-\text{raw}^2/2\sigma^2}$); pooled over bands, shared photons preserve lucky agreement more often than chance predicts.
- Only the **bright** slice grows a tail at large raw offsets, where chance predicts zero. Right: worseners sit right of the improved because the selection forces it (res > raw truncates the floor distribution from below); the truncated-floor prediction is 21.6 mas median versus 17.8 observed. Only the excess beyond ~ 40 mas escapes any chance model

For one source, per band b : fit the classical centroid, form $\text{raw}_b = \text{pos}_b - \text{pos}_{\text{VIS}}$; the head (v10) predicts a correction c_b from one shared representation; $\text{resid}_b = \text{raw}_b - c_b$. “Strongly worsened in b ” = $|\text{resid}_b| > 1.5 |\text{raw}_b|$ and worse by > 10 mas. Rate: 5.6% of bright measurements.

- **Independent between bands:** the centroid-fit noise of each band image (different exposures, different photons). If this alone drove worsening, hits on the same source would be independent coin flips: $P(3 \text{ hits}) \approx 0.056^3 \times 10 \text{ triples} \times 3185 \text{ sources} \approx 5$.
- **Shared for one source:** (1) the VIS reference position, subtracted from every band; (2) the local WCS terms (Rubin bands share Rubin’s solution, NISP its own); (3) the source itself (proper motion, colour gradients); (4) the head’s shared representation, feeding all nine corrections.

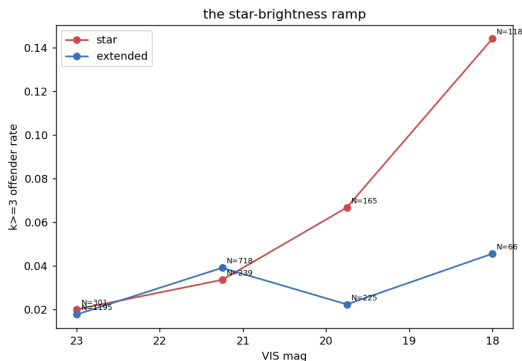
strongly worsened in $\geq k$ Rubin bands	observed	chance (permutation null)
$k \geq 2$	181	68 ± 7
$k \geq 3$	113	5 ± 2
$k \geq 4$	61	0.2
$k \geq 5$	23	0.0

The null shuffles flags within (band, raw offset, S/N) classes, preserving all rates and the VIS-induced raw correlations (hence chance $k \geq 2$ is 68, not 0). Robust to the cuts (S/N>10: 200 vs 8; all nine bands: 153 vs 12) and to the flag definition: the excess exists at every threshold. Tightening just cleans the list of chance interlopers (fiducial: 113 flagged, only ~ 6 innocent), making it clean enough to study source by source.

Stage 1 established a source-attached, band-shared cause, not yet necessarily the head. Four things are shared; each gets a measurement:

- **(1) The VIS reference?** A bad pos_{VIS} would worsen all bands at once, a plain measurement error. Gaia says no: for the offender stars the VIS labels sit **7.0 mas** from the PM-propagated Gaia position, better than the survey average.
- **(2) The WCS?** A solution error is smooth in sky position: it cannot single out one star and spare its neighbours. Offenders show **no sky clustering** (nearest-offender distance $62''$ vs $66'' \pm 5''$ for random draws), and the failure **persists in NISP** (59% of offenders worsened in all NISP bands), whose astrometric solution is independent of Rubin's.
- **(3) The source itself?** Proper motion correlates with nothing ($\rho = 0.01$ against residuals); high-PM stars are actually the head's success cases (it bridges the epochs).
- **(4) The head's representation** carries positive fingerprints, not just survival by elimination: corrections **larger than the disagreement they correct**, directions coherent across all nine bands and **both instruments** (one wrong belief applied everywhere), and Gaia places the head far from truth where every classical measurement of the same stars is at the floor (numbers on the next slides).

The two-stage verdict: repetition proves the cause is source-attached and shared; elimination plus fingerprints prove it is the head. The 113 are 0.3% of the catalog, identifiable one by one. (All measurements on this slide: v10 production head.)

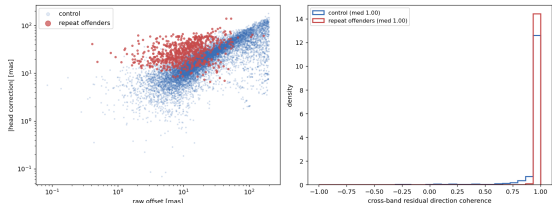


Offender rate (**v10 production head**) for stars ramps 2% → **14%** from VIS 24 to VIS 17; extended sources stay flat.

Population: the 3,185 sources with ≥ 2 bright (S/N>20) Rubin bands, the only ones the k statistic can score, out of 37,796 in the catalog; N labels are the per-bin denominators of that eligible population, and rate $\times$ N summed over bins and classes recovers the 113.

Further mechanisms tested and **ruled out** (beyond the WCS and proper motion of the previous slide):

- tile-edge context (paired same-source test: ± 0.1 mas)
- companions within $2''$ (13% vs 20% control)
- VIS saturation (onset ≈ 17 , masked not clipped; offender cores pristine)

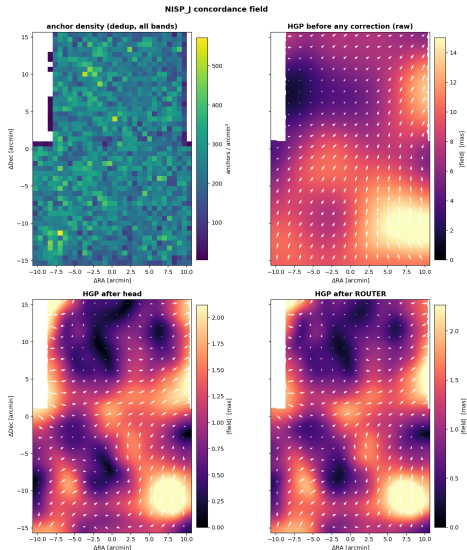


Against PM-propagated Gaia DR3 (epoch 2024.83), per star:

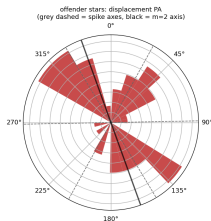
	VIS label	head (v10)
offender stars ($k \geq 3$)	7.0 mas	22.9 mas
all Gaia stars	9.5 mas	12.2 mas

Left: offenders receive corrections **larger than the raw disagreement** ($|c| \sim 28$ mas vs raw ~ 11 mas), coherently in every band: the head asserts a center the classical centroids contradict.

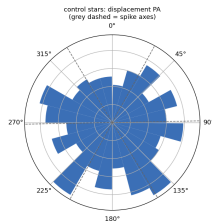
July reading: a brightness-dependent divergence between the Gaussian-fit convention and the head's center. Interim mitigation: an uncertainty-weighted **router** between head and classical positions (bright stars back at the classical 5 mas floor). *The mechanism was found in August — slides ahead.*



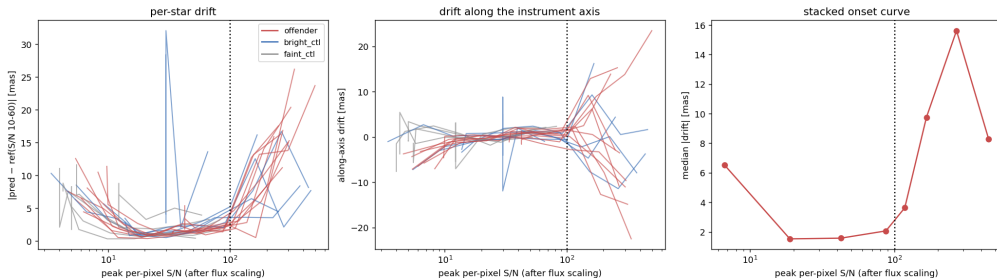
- Refit the field (HGP super-tight, PINN) on the router's residuals (v10-era archives): median change **0.16 mas**, an order of magnitude below the field amplitude. The robust fit had already absorbed the 113 outliers; Figure 10 and the FITS products stand.
- The ~ 15 mas lobe at lower right is **real**: coherent in 8 of 9 bands relative to VIS, normal anchor coverage, both solvers agree, and it sits within MER tile TILE102043549. Eight rulers agree and one does not: a localized feature of the **VIS solution**.
- By construction no VIS-trained correction can remove it; the released raw-field FITS is precisely the map that carries it.



v10 archive (ECDFS)

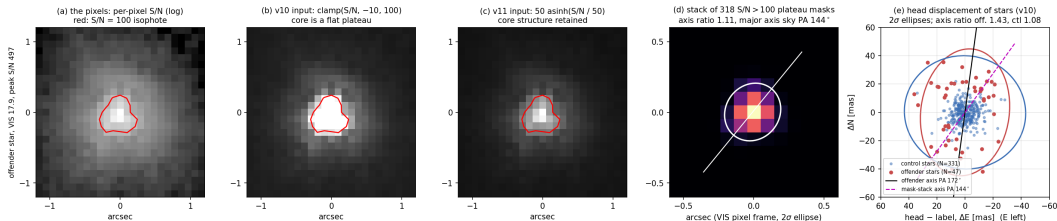


- Offender displacements are **elongated along a single sky axis** ($PA \approx 160^\circ$, both signs, no net direction); controls are isotropic.
- The axis **rotates with the Euclid instrument** between ECDFS and EDF-S (spike phases $28.9^\circ \rightarrow 49.7^\circ$).
- Published VIS geometry (strut angles + four-corner CCD readout) predicts the readout axis **1.9°** from the measured one, with zero tuned parameters — in both fields, one parity.
- Yet a signed flux stack shows **no trailed light** (CTI ruled out), and rotating a star's own pixels rotates its displacement: the cause lives in the images as the network sees them.

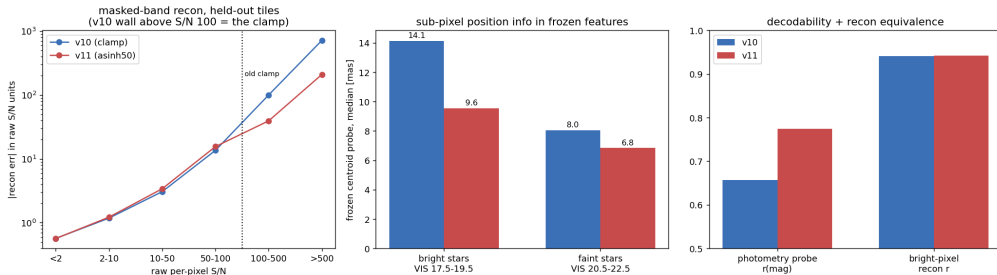


causal interventions on the v10 production model

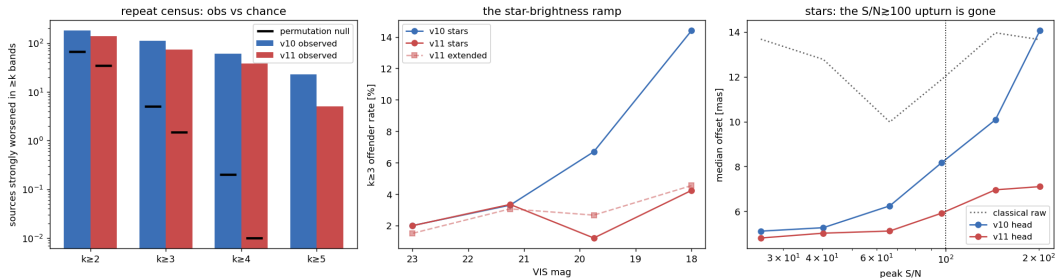
- Every input enters the foundation as `(image/rms).clamp(-10, 100)` — unchanged since v4. Above per-pixel S/N 100 a star's core is a **featureless plateau**; the reconstruction targets are clamped identically, so **no loss ever saw the missing information**.
- Causal test: scale a real star's flux $\times 50$ — the head places the *same star* to 1–2 mas for peak S/N 10–90, and the displacement **switches on exactly above $S/N \approx 100$** (a threshold, not a trend), reaching the archived 11–26 mas at native flux.
- Rotating the star's local image 180° flips the displacement (median $\cos = +0.91$): each star's error is deterministic, written in its own pixels.



- Above S/N 100 the core is a plateau, so the head infers the center from the **plateau boundary** and wings. The VIS PSF is slightly elongated in the instrument frame, so that boundary (red contour, a) is an ellipse: per star a ragged blob, but the **stack of 318 plateau masks** (d) has axis ratio 1.11 with major axis PA 144°, matching the offender displacement axis (e; $m=2$ axis $160^\circ \pm 9^\circ$, covariance axis 172°). Panels b and c show the same star as the network saw it before and after the fix.
- An elongated boundary constrains the centroid well across its short axis and **weakly along its long axis**: errors along one axis, both signs, no net drift. A mildly elliptical boundary gives a strongly anisotropic error. No trailed light in the flux stack (CTI ruled out).
- Controls are isotropic because they are **not clamped**: stars below S/N 100 keep their cores, galaxies never reach it. Every clamped star draws from the same mechanism; the flag selects the large-amplitude tail. Under v11 the elongation is still in the images, but with the core restored the head stops leaning on the boundary and the axis vanishes. Per-star sign untested.



- `clamp(-10,100) → 50*asinh(x/50)` in stems *and* reconstruction targets; foundation retrained from scratch, identical recipe (`jaisp_v11_q1_soft`).
- Gate: reconstruction parity in bulk (bright-pixel r 0.9412 vs 0.9419); the v10 error **wall above S/N 100** is gone (left); frozen centroid probe on bright stars **14.1 → 9.6 mas** (middle); photometry decodability up.
- The position head is retrained **unchanged** — no side-channel, no reweighting. The architecture was never the problem; its inputs were.



- The star-brightness ramp is **flat** (2–4% at all magnitudes, indistinguishable from galaxies; was 2 → 14.4%).
- The $S/N \gtrsim 100$ upturn is **reversed**: the v11 head sits at 5–7 mas where classical raw is 10–14 — it now adds value on bright stars. General Gaia stars: head at label parity (10.8 vs 9.5 mas).
- Faint-end headline untouched (raw 47.9 → 13.3 mas, 91% improved — identical to v10). Under v11 the router no longer improves any regime: the interim mitigation is **retired by the cure**.

Decomposition of the 113 repeat offenders:

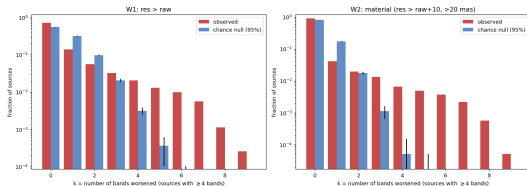
~46 clamp-driven bright stars	cured by v11	(the ramp, the upturn, the axis)
~67 source-specific cases	survive both foundations	(the new open question)

The survivors: the *same sources* under two independently trained foundations, same signature (raw ~ 10 mas, coherent ~ 30 mas head displacement; Gaia still against the head, 29.5 vs 9.6 mas for the 8 Gaia stars among them). Not brightness-selected, not resolved blends (12% vs 20% control), not extended.

- Cause is outside the foundation: in the data or labels of these specific sources, or in what both trainings share (label pipeline, tiles). Suspects: **sub-arcsecond pairs below MER's deblending limit**; convention-ambiguous profiles with unstable training labels.
- 0.2% of the catalog; the uncertainty router covered them in the meantime. **Resolved in September — next slides.**

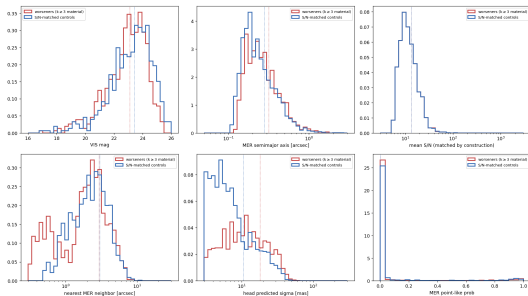
September: the survivors are a real population — and it was there in v10 too

15/21

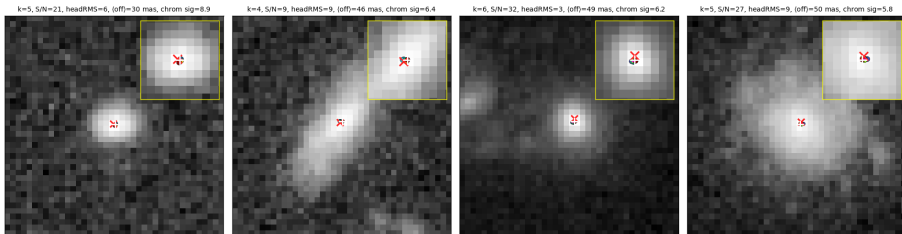


v11 plain head, 790 tiles, 37.9k sources (io/31)

- Count in how many bands a source is *worsened by more than 10_{mas}* (head residual > raw+10 mas and > 20 mas; “any worsening” = residual > raw). The $k \leq 2$ bins match the chance null built from empirical per-(band, S/N) rates: the bulk of the 8.9% worsened measurements is luck.
- At $k \geq 3$: **604 sources (1.6%)** vs 23 ± 5 by **chance** ($\times 26$; $\times 400$ at $k \geq 4$). Within a source the residual vector points the same way in every band (median pairwise cos = 0.997): one deterministic wrong position per source.
- Isotropic across sources, no tile clustering, no sky pattern: nothing instrumental survives in v11. The cause lives in each source’s own pixels.
- Rerun on v10: **622 sources, 80% the same objects**. Hidden at S/N ≈ 13 , where one worsened band looks like chance; the bright-tail selection of July was dominated by the clamp instead.

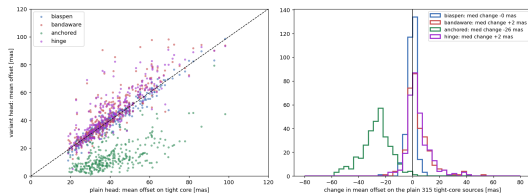


- vs S/N-matched controls: galaxies in both (point-like 1%), but **brighter (VIS 23.1 vs 23.4) and larger (0.32'' vs 0.28'')** at the same S/N; $\times 1.7$ excess of sub-arcsecond MER companions (7.6% vs 4.5% within 1''). The head's own σ is elevated (18 vs 11 mas): it partially knows.
- Against the VIS light itself (peak, and 0.75'' flux centroid): the head sits **2–2.6 $\times$ farther than the label** on worseners, isolated and paired alike (50 vs 23 mas to the peak); on controls head and label are equivalent. Band-to-band the head is the tighter estimator everywhere (13 vs 31 mas RMS).
- Reading: **precise but biased** — one reproducible ~ 40 mas offset per source matching no natural photometric center; on chromatic sources it lies along the color axis, toward the blue/clumpy light. The **tight core of 315 sources (0.8%)**, 89% persistent across v10/v11, is the true model-bias population.



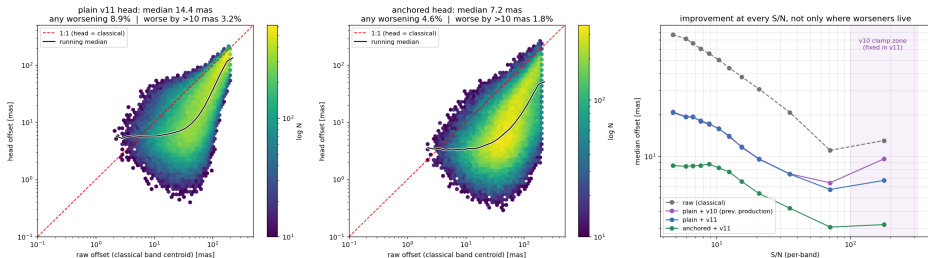
VIS stamps 4'' wide, 1'' inset. Dots: classical per-band centroids colored by wavelength (*u* blue → *H* red); magenta: chromatic axis; red ×: plain-head position; white +: VIS label. Tight-core members, plain v11 head (io/31).

- Everything here is **sub-pixel**: both estimators put the source “at the center” to within half a VIS pixel; the disagreement is tens of mas.
- The classical centroids drift **systematically with wavelength**: the nine dots trace a short ellipse along the magenta axis. A structured galaxy has no unique center; each band and each rule defines one.
- The head’s red cross sits in **one fixed spot, tight across all nine bands**, displaced 30–50 mas from the label, typically toward the blue, clumpy end of the axis. Precise, repeatable, and off the label’s definition: that is the convention bias.
- Neither position is “wrong”. The head learned a morphology- and color-weighted center; the label is the VIS Gaussian-weighted centroid we score against. The anchored head removes the disagreement by **computing the label’s rule** in its forward pass.

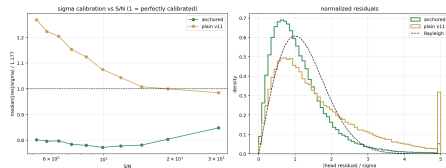
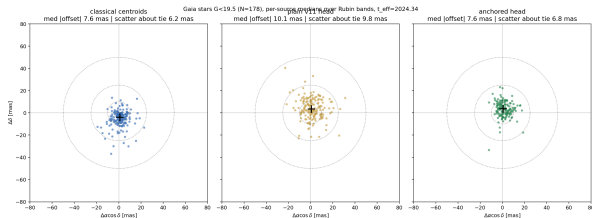


head	overall med	faint med	coherent ($k \geq 3$, null)	tight core	$\cap$ plain 315	control RMS
plain	14.4	18.8	604 (22)	315	—	9.9
biaspen (loss)	15.1	19.7	542 (18)	297	260	10.5
bandaware (arch.)	9.7	10.9	1113 (89)	662	283	6.8
hinge (loss)	9.6	10.8	1061 (82)	639	282	6.9
anchored	7.2	8.6	235 (2)	37	6	7.7

- Identical rows scored for every head, no worsener ever a training signal (medians in mas). **Anchored**: an iterated Gaussian-weighted VIS centroid computed in the forward pass, with its photon-noise σ ; the network adds only a residual bounded at a few $\times$ that noise. Tight core **315 $\rightarrow$ 37 (6 of the original survive)**, every global metric better at once. Three loss-based fixes failed (NLL, biaspen, hinge): the failure is representational, not an optimization artifact.



- Anchored sits below plain at **every** raw offset and every quantile: p50 **14.4** $\rightarrow$ **7.2** mas, p90 79 $\rightarrow$ 45, faint tercile 19.7 $\rightarrow$ 8.5, bright 9.4 $\rightarrow$ 5.3. Any worsening 8.9 $\rightarrow$ 4.6% of measurements; worsened by more than 10 mas 3.2 $\rightarrow$ 1.8%; improves on raw for 95.4%.
- The faint end, where no worsener was ever flagged, improves the most: there the bound is open and the learned residual does the work. On bright stars the anchor pins the head to the classical floor.
- Right: the v10 upturn above S/N 100 (clamp) vs v11; the anchored gain on top is head design, not foundation version.
- Adoption gates passed (2026-08-31): **EDF-S with no retraining** median 6.4 mas (plain 13.9); **patch-disjoint** retrain 6.9 mas vs 9.2 random-split — no overlap leakage. Adopted as production 2026-09-01; retrained on v11 detector labels 2026-09-06 (latent_position_v11_anchored_v11labels), the archive behind paper Figs. 7, 8, 11.



- Gaia DR3, PM-propagated; the Q1 coadds' effective epoch re-derived as $t_{\text{eff}} = 2024.34$ by all three estimators (the 2024.83 value belonged to the older coadds).
- Bright stars ($G < 19.5$): **anchored = classical**, 7.6 mas median offset for both, scatter about the tie 6.8 vs 6.2. The plain head pays a visible Gaia penalty (10.1 / 9.8): its convention bias is real to an external ruler. Amortization now costs nothing on stars.
- The 7.8 mas Dec split between the classical tie (-4.1) and the head tie ($+3.7$) is the concordance field (common component 6.9 mas), not a pathology: classical Rubin centroids live in the Rubin frame, the heads in the VIS convention, Gaia sits between.
- σ_{pred} : anchored is **uniformly conservative, factor 0.79 flat in S/N** — one global rescale calibrates it (90% coverage 91%). Plain σ was miscalibrated in shape with a heavy tail (90% coverage 74%). The router has at most ~ 0.5 mas left to collect: retired.

- Above the 1:1 line sit two populations: mostly benign regression to the corrected floor, plus real failures concentrated in bright point sources — diagnosed by **repetition across bands**, not by any average metric.
- The mechanism was one line of preprocessing: a hard per-pixel S/N clamp that erased bright-star cores — invisible to every loss curve because the targets were clamped too. **Audit your input transforms; stratify metrics by regime.**
- Correlational forensics narrowed the suspects; **causal interventions** (flux sweep, rotation) convicted. The fix is soft compression + retrain (v11): ramp flat, upturn reversed, faint-end headline untouched, detection deeper.
- The interim uncertainty router did its job and is retired by the cure. The 67 survivors were the visible tip of a **1.6% extended-galaxy population** where the learned center disagrees reproducibly with the classical convention — present in v10 too, hidden at $S/N \approx 13$, invisible to loss-based fixes.
- The cure for that one is architectural: the **anchored head** computes the convention centroid in its forward pass and learns only a bounded residual. Tight core $315 \rightarrow 37$, medians halved at every S/N, Gaia parity on bright stars, calibrated σ ; gates passed on EDF-S and patch-disjoint. Production since 2026-09-01.
- The paper sentence, updated: the head improves astrometry where classical methods are noise limited, and **matches them** where they are systematics limited — a single estimator for both regimes.